Sex-equalized high-BMI exposure and potentially avoidable MASLD/NAFLD liver burden

GBD 2023 counterfactual analysis across Southeast Asia, East Asia, and Oceania

1 Evidence base

GBD 2023 estimates for MASLD/NAFLD-related cirrhosis and liver cancer burden.

34 locations

Adults 20+

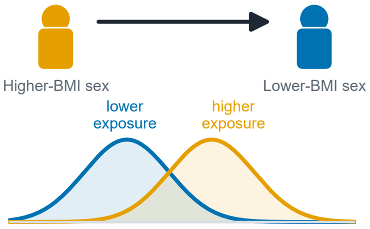
1990-2023

WPP 2024

High-BMI exposure, attributable burden, SDI context, and future person-years.

2 Counterfactual

Higher-BMI sex assigned the lower-BMI sex exposure distribution within each location-year-age cell.



Scenario A: sex-equalized high BMI

Scenario B: TMREL upper bound

Nested deterministic layers; not additive

3 2023 burden

Potentially avoidable MASLD/NAFLD-related liver burden under Scenario A.

2,797

potentially avoidable DALYs

107 deaths

0.40% of total DALY burden

11.3% of observed high-BMI DALYs

Effect is small but measurable

4 Risk priority

High BMI was not the dominant regional MASLD/NAFLD-related risk signal.

Risk	Locations
High FPG	27/34
High BMI	3/34
Smoking	4/34

2050 internal DALY-rate scenario

Scenario	Value
Baseline	47.1
Scenario A	46.2
Scenario B	39.3

Scenario A: -1.9% | TMREL upper bound: -16.6%

Glycaemic-risk prevention remains central

Scenario A assigns the higher-BMI sex the lower-BMI sex exposure distribution within each location-year-age cell.
Scenario B is a high-BMI-at-TMREL upper-bound comparator; the scenarios are nested and not additive.
Projection rates are internal scenarios, not IHME official forecasts.